

CS 105: Department Introductory Course on Discrete Structures

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Lecture 05 – Induction, WOP and Strong Induction

Recap

Till now

- ▶ What are discrete structures?
- ▶ Course outline
- ▶ Chapter 1: Proofs and structures
 - ▶ Propositions: statements that can be assigned a truth value
 - ▶ Predicates: propositions with variables
 - ▶ Quantifiers
 - ▶ Theorems and different types of proofs

Last week

- ▶ Induction
- ▶ Well-ordering principle
- ▶ Strong Induction

Quiz 1: Aug 22nd, 8.15am, Venue to be announced.

Induction examples

An odd number of students stand at mutually distinct distances. At the same time, each student throws a paper rocket at their nearest neighbour, hitting this person. Use mathematical induction, to show that there is at least one survivor, i.e., at least one student who is not hit by a rocket!

How to show this? What is $P(n)$?

- ▶ $P(n)$ be the statement that there is a survivor whenever $2n + 1$ students stand at distinct mutual distances and each student throws a rocket at their nearest neighbour. Show that $P(n)$ is true for all positive integers n .
- ▶ Base case: $n=1$.
- ▶ Assume for groups of $2k + 1$ students. Now consider a group of $2k + 3$ students.
- ▶ Consider the closest pair A-B and divide into two cases based on whether someone threw rocket at them.

Formal Writing

Qn: Prove that $\forall n \in \mathbb{N}, n \geq 1, 1 + 2 + \dots + n = \frac{n(n+1)}{2}$

Proof by Induction

1. **Base case** $n = 1$, $\text{lhs} = 1 = \frac{1 \cdot 2}{2} = \text{rhs}$. Hence holds.
2. **Ind Hyp:** Assume statement for some k (or for all numbers from $1 \dots k$): $1 + 2 + \dots + k = \frac{k(k+1)}{2}$
3. **Ind Step:** at $k + 1$, we need to show
$$1 + 2 + \dots + (k + 1) = \frac{(k+1)(k+2)}{2}$$
$$\text{lhs} = 1 + 2 + \dots + k + (k + 1) = (1 + 2 + \dots + k) + (k + 1)$$
$$= \frac{k(k+1)}{2} + (k + 1) \text{ (By Induction Hypothesis)}$$
$$= \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+2)(k+1)}{2} = \text{rhs}.$$
4. Hence by induction we can conclude for all $n \in \mathbb{N}, n \geq 1$.

Formal Writing

Qn: Prove that $\forall n \in \mathbb{N}, n \geq 1, 1 + 2 + \dots + n = \frac{n(n+1)}{2}$

Proof by WOP

Any non-empty set of non-negative integers has a smallest element.

1. Suppose Not. i.e., the negation of the statement holds.
2. There exists some k for which $1 + 2 \dots k \neq \frac{k(k+1)}{2}$.
3. Let S be set of all such counter-examples, i.e.,
$$S = \{ \ell \in \mathbb{N} \mid 1 + 2 \dots \ell \neq \frac{\ell(\ell+1)}{2} \}$$
4. S is non-empty! Why? Because by above, $k \in S$.
5. By WOP, S must have a smallest element, let it be $k' \in S$.
6. $k' \neq 1$, so $k' - 1$ exists. Because for $n = 1$, we use Base case!
7. At $k' - 1 < k'$, statement is true, i.e.,
$$1 + 2 \dots (k' - 1) = \frac{(k'-1)(k')}{2}.$$
8. But now, $1 + 2 \dots (k' - 1) + k' = \frac{(k'-1)(k')}{2} + k' = \frac{k'(k'+1)}{2}$.
9. Implies $k' \notin S$. A contradiction.

Application of WOP to prove theorems (Last class)

- Proving one part of the **fundamental theorem of arithmetic**.

Theorem: Any integer > 1 can be written as a product of (one or more) primes

Proof by contradiction **using WOP!**:

- ▶ Let S be the set of all integers greater than 1 that cannot be written as a product of (one or more) primes.
- ▶ If S is non-empty, there is a least element in it by WOP.
- ▶ Call this least number n . First, n can't be a prime (why?).
- ▶ So $n = a \cdot b$, where $n > a, b > 1$.
- ▶ Since a and b are smaller than the smallest number in S , they can be written as a product of (one or more) primes.
- ▶ Let $a = p_1 \dots p_k$ and $b = q_1 \dots q_l$, for $k, l \geq 1$. But then $n = p_1 \dots p_k \cdot q_1 \dots q_l$, which is a contradiction. □

Qn: How do you show uniqueness?

Fundamental theorem of arithmetic (Uniqueness)

*Any integer > 1 can be written as a **unique** product of one or more primes*

How do we show uniqueness? **Proof by contradiction!**

1. Let $S \neq \emptyset$ be the set integers > 1 that can be written has product of one or more primes in two ways.
2. By **WOP**, let s be the smallest integer in S .
3. Then $s = p_1 \dots p_m = q_1 \dots q_n$ where each p_i is distinct from each q_j .
4. **Without loss of generality**, assume $p_1 < q_1$.
5. Then, $s = p_1 \cdot P = q_1 \cdot Q$ for some $Q < P$.
6. $s - p_1 Q = (q_1 - p_1)Q = p_1(P - Q) < s$ which implies $(q_1 - p_1), Q < s$.
7. But by 2., Q and $(q_1 - p_1)$ must have a unique prime factorization, so p_1 must occur in it.
 - ▶ If p_1 occurs in factorization of Q , then $p_1 = q_j$ so violates 3.
 - ▶ If p_1 occurs in factorization of $q_1 - p_1$ then p_1 must divide $q_1 - p_1$ and hence also q_1 , which is impossible since p_1, q_1 are primes.
8. Hence we have a contradiction, and we conclude $S = \emptyset$. $\square$

Strong Induction

Strong Induction

Let $P(n)$ be a property of non-negative integers. If

- ▶ $P(0)$ is true (Base case)
- ▶ for all $k \geq 0$, $(P(0) \wedge P(1) \wedge \dots \wedge P(k)) \implies P(k+1)$
then $P(n)$ is true for all $n \in \mathbb{N}$. (Induction Step)

Induction

Let $P(n)$ be a property of non-negative integers. If

- ▶ $P(0)$ is true (Base case)
- ▶ for all $k \geq 0$, $P(k) \implies P(k+1)$ (Induction step)

then $P(n)$ is true for all $n \in \mathbb{N}$.

Theorem: Strong Induction iff Induction iff WOP

Another exercise by Strong Induction

Quotient-Remainder Theorem

For any $m, n \in \mathbb{N}$, $m \neq 0$, there exists a **unique** quotient q and remainder r ($q, r \in \mathbb{N}$), such that

$$n = q \cdot m + r, \quad 0 \leq r < m$$

1. Fix any $m > 0$, we use strong induction on n .
2. Base cases: $n \in \{0, \dots, m-1\}$. What should q and r be?
3. Induction step: We prove for all $k \geq m$,
 - 3.1 (ind hyp): if $\forall n \in \mathbb{N}$, $n \leq k$, $\exists q, r$ s.t. $n = qm + r, 0 \leq r < m$.
 - 3.2 (to prove): $\exists q^*, r^*$ s.t., $k+1 = q^*m + r^*$ for $0 \leq r^* < m$.
4. Take $k' = (k+1) - m = k - (m-1) \leq k$. i.e., $k' \leq k$, so we can apply ind hyp on k' .
5. By Ind Hyp, $k' = q'm + r'$ for some q' , $0 \leq r' < m$.
6. Now to prove 3.2, we choose $q^* = q' + 1, r^* = r'$
7. Then, $q^*m + r^* = (q' + 1)m + r' = q'm + r' + m = k' + m = ((k+1) - m) + m = k+1$. □

Qns: Show uniqueness (H.W). Also, what if $m, n \in \mathbb{Z}, m \neq 0$?